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An Outstanding Record of Paleozoic Siliciclastic Sedimentation and Unconformities in the Subsurface of Saudi Arabia

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Abstract

This study is focused on seven different wells drilled in the Rub' al-Khali Basin (southeastern Arabian Plate) and shows one of the few available integrations of palynological and sedimentological data across its Paleozoic stratigraphy, with implications for a better knowledge of both the geological history and the petroleum geology of this basin. Middle Cambrian strata, partially assigned to fluvial and coastal environments from the Siq and Burj formations respectively, represent the oldest intervals observed in our wells. Above, the upper Cambrian to Middle Ordovician Saq Formation is identified in all the wells except in those from the northern sector of the studied area. While the northern sector is interpreted as a local paleo-high that experienced little or no sedimentation during large parts of the Paleozoic, a core cut in the upper section of the Saq Formation is interpreted as partly deposited in a fluvial setting. A ~15 Ma unconformity is inferred between the Saq Formation and the upper members of the Qasim Formation (Middle to Upper Ordovician). The latter unit includes cores assigned to marine offshore settings. Also, parts of the uppermost interval of the Qasim Formation are likely eroded by the latest Ordovician glaciation. Uppermost Ordovician glacial strata from the Sarah Formation are present in the southern and central sectors of the studied area. Palynological assemblages indicate that at least part of the lowermost Sarah Formation belonged to relatively distal lacustrine settings. However, a core cut in the upper part of this unit exhibits intensely deformed strata probably related to glacial ice dynamics impacting fluvial or marine deposits. Above the Sarah Formation, a relatively thin section of the lower to middle Silurian Qalibah Formation is only found in three of the studied wells, a situation that is interpreted as due to major erosion during the formation of the Hercynian Unconformity (late Carboniferous). An increased organic richness, often documented elsewhere in the lowermost Qalibah Formation across the Arabian Plate, may still be present in these three wells. No Devonian strata are found in the studied area; instead, the glaciogenic Juwayl Formation (upper Carboniferous to Lower Permian) directly overlies the Hercynian Unconformity and shows large thickness differences across our wells. The latter is likely due to differences in the available accommodation space that was subsequently filled by sediments resulting from the shaping of the topography by glacial ice advances and retreats. The overlying Nuayyim Formation (Lower Permian) exhibits comparatively persistent thickness across the studied area, to include the northern sector, and suggests a period of structural quiescence, in terms of constant subsidence and fill of the available accommodation space,

during its deposition. Finally, 2 cores from the upper section of the Nuayyim Formation include mixed clastic-carbonate sediments deposited in coastal to shallow-marine environments that evolved into subaerial paleosols probably due to an increased climatic aridity by the end of the Lower Permian. A corollary is that, unlike the continental aeolian systems commonly interpreted across sectors of the eastern Arabian Plate, our inferences point to the absence of aeolian dunes in our wells but the possibility that a dynamic interaction of coastal and marine systems existed in parts of the Rub' al-Khali Basin by the early Permian.

Introduction

Large parts of the Paleozoic sedimentary record across the Arabian Plate are dominated by siliciclastic units and numerous publications reveal that these can be grouped into stratigraphic packages bounded by unconformities that incorporate different degrees of non-deposition and/or erosion (Stump et al., 1995; Wender et al., 1998; Al-Hajri and Owens, 2000; Sharland et al. 2001; Al-Husseini and Matthews, 2006), e.g., the upper Carboniferous Hercynian Unconformity and the middle Permian pre-Khuff Unconformity that constitute, respectively, the bottom and top bounding surfaces of the Unayzah Group. The dominance of siliciclastic strata is ultimately related to plate tectonics, i.e., the Arabian Plate belonged to the northern margin of Gondwana during large parts of the Paleozoic and its drifting near high to mid-latitudes broadly promoted low relative sea level and the availability of vast continental areas subject to protracted erosion and increased siliciclastic supply (e.g., Sharland et al., 2001; Haq and Al-Qahtani, 2005; Ruban et al., 2007). Under these conditions, regional trends such as the gross depositional environments of the major stratigraphic units and their biostratigraphy are reasonably constrained by a wealth of studies from surface outcrops (see e.g., Sharland et al., 2001 and references therein). The published studies focused on the subsurface are, albeit less abundant, still numerous due to the notorious interest in petroleum geology (e.g., Evans et al., 1997; Craigie et al., 2016; Stewart, 2016; Miller et al., 2017). Regardless, a main source of uncertainty using subsurface data comes from the fact that the Paleozoic basins of the Arabian Plate are massively extensive and this situation leads to interpretations of sedimentological and biostratigraphic patterns that may become vague when creating basin-wide extrapolations, especially in underexplored areas. In this work, we present new results of the Paleozoic stratigraphy from seven different wells located in the Rub' al-Khali (RAK) Basin (southeastern Arabian Plate). The area of study is intentionally restricted in space so that uncertainties related to data distribution are reduced. The ultimate goal is to revise the stratigraphic trends of the Cambrian to Permian major units below the carbonate strata of the Khuff Formation (Fm). That is, we aim to refine the sedimentological, stratigraphic, and palynological patterns in this basin and the implication is that a detailed understanding of these patterns will lead to a better knowledge of its geology

Regional Geology

The majority of the plate reconstructions indicate that the Arabian Plate was part of the Tethyan margin of Gondwana that drifted between 30-60° south paleolatitudes during most of the Paleozoic (Sharland et al., 2011; Ruban et al., 2007). In the southeastern sector of the Arabian Plate, the RAK Basin was one of its largest depocenters that at present includes a broad synform of a relatively complete Phanerozoic succession that unconformably overlies Precambrian sedimentary, metamorphic, and igneous basement (Stewart, 2016). Only along the edges of the RAK Basin near its westernmost sectors, the Precambrian basement is exposed in the surface and forms part of the so-called Arabian Shield that widely extends westward across the Arabian Plate. Regional studies indicate that this basement is composed of oceanic and continental terrains that successively collided during the Neoproterozoic and created separating shear zones now broadly oriented N-S, with local variations between NW-SE and NE-SW (see Stewart, 2016 and references therein). Similarly-oriented structural trends are interpreted in the basement across the RAK Basin and, in practice, some of them are considered major fault zones that could bound local depocenters

and relative highs subject to differential sedimentation and/or erosion at least during parts of the Paleozoic. In addition to being affected by basin topography, the Paleozoic stratigraphy of the RAK Basin was strongly influenced by periods of major erosion such as that during the late Carboniferous, with the resulting formation of the Hercynian Unconformity. Such an erosional event is considered to have produced major differences in the thickness of the Cambrian to Carboniferous succession in the RAK Basin (e.g., [Sharland et al., 2001](#)).

Materials and Methods

The materials studied in this project derive from seven different wells drilled in the RAK Basin. The wells include a wealth of legacy palynological information that was reviewed in this project and integrated with the data generated from new palynological samples from the same wells. The resulting Paleozoic palynozones from the wells were used for age dating, to assign formation names to the different stratigraphic intervals and, ultimately, to produce chronostratigraphic well correlations. Also, the sedimentology of a total of 630 ft of conventional core from parts of the stratigraphy of the studied wells was described at 1:20 scale, to include the main lithologies, grain size, major components, composition, sedimentary structures, diagenetic features, and vertical stacking patterns. Based on the latter, depositional conditions were interpreted for each core and this information was integrated with their stratigraphic assignment to broadly predict the spatial distribution of both sedimentary environments and elements of petroleum systems.

Results and Implications

Middle Cambrian

The lowermost interval of the studied stratigraphy is assigned to the middle Cambrian Siq Fm and this is present in Well-1. A core cut at the bottom of Well-1 is interpreted to belong to this unit and consists of fine-to medium-grained, moderately-to poorly-sorted sandstones that are intensely folded and fractured. The core is tentatively assigned to a fluvial system and its intense levels of deformation, compaction, and cementation are responsible for the firm tightness of the sediments. Above the Siq Fm, the middle Cambrian Burj Fm is inferred in Well-1. This unit is 650 ft thick and its upper part has a core that shows red and green claystones and a few layers of light gray, fine-grained, massive to horizontal-bedded sandstones, some of them with erosional bases. The claystones may show microbial laminations and associated desiccation cracks. Together, these lithologies suggest the existence of a coastal plain cut by fluvial channels at least in the upper part of the Burj Fm in this location. Well-2 is 150 km to the ESE of Well-1 and its lower half is assigned only a broad pre-Saq Fm interpretation due to the very lean or barren palynology from cuttings and core samples. Well-2, however, includes three different cores dominated by black, massive claystones along with intervals of very fine-grained matrix (clay and silt) conglomerates with abundant pebble-sized floating lithoclasts. A high degree of fracturing is also seen in these sediments. An interpretation is that these cores could have been deposited in an offshore setting broadly during parts of the Neoproterozoic to middle Cambrian, possibly not far from a slope feeder-channel system based on the mutual presence of claystones and conglomerates. Regardless, the intense compaction and fracturing of these sediments are probably evidence of no potential open pore spaces at least within this interval in Well-2.

Upper Cambrian to Middle Ordovician

The palynological observations indicate the presence of Lower to Middle Ordovician strata in Well-3, 4, 5, and 7 that are attributed to the Sajir Member, i.e., the upper part of the Saq Fm. Only Well-7, in the southernmost sector of the study area, includes a core that reveals m-thick bedsets of erosive-based, normally-graded, low angle to trough cross-bedded sandstones (medium to coarse in grain size and moderately-sorted), which are assigned deposition in a fluvial setting. The aforementioned interpretation

that the middle Cambrian Siq and Burj formations are present in the northern sectors of the study area leads to the notion that the overlying Saq Fm (i.e., the Risha and the Sajir members) may occur in most of the RAK Basin. Importantly, compaction and cementation of the cored sediments from the Sajir Member in Well-7 are not very intense so that some open pore spaces may be preserved, a situation that otherwise elevates the potential of the Saq Fm in the RAK Basin.

Middle to Upper Ordovician

Our palynology reveals that only a section of the upper members of the Qasim Fm (i.e., Ra'an and Quwarah) (350-650 ft thick) is present in the central and southern sectors of the studied area (Well-3 to 7). The lower members of the Qasim Fm (i.e., Hanadir and Kahfah) are not found in any of the studied wells, whereas the entire Qasim Fm (Middle to Upper Ordovician) is absent in the wells from the northern sector (Well-1 and 2). The absence of the Hanadir and Kahfah members in these wells must have been related to erosional and/or non-depositional events before sedimentation of the Ra'an Member that created an unconformity of at least ~15 Ma above the Saq Fm. Also, the complete absence of the Qasim Fm in the northern sector is interpreted as this sector being a depositional paleo-high during large parts of the Paleozoic. Regardless, the studied palynological assemblages indicate that the transition from the Ra'an into the overlying Quwarah members is captured in a core cut in Well-3. This mostly shows sandy siltstones with some shelly fragments that together are attributed deposition in an outer shelf. Another core was cut near the top of the Quwarah Member in Well-6 that exhibits a section of micaceous siltstones and a few layers of fine-grained sandstones, which are likewise interpreted as deposition in an outer shelf. That is, the cored lithologies suggest only seal potential in the upper members of the Qasim Fm probably across the studied areas of the RAK Basin. Finally, our well correlations reveal that the uppermost section of the Quwarah Member is not present in sectors of the RAK Basin, probably due to differential erosion and incision related to the latest Ordovician (Hirnantian) glaciation.

Uppermost Ordovician

Our palynology and well correlations show a 730-1160 ft-thick interval that is assigned to the uppermost Ordovician Sarah Fm in the wells from the southern and central sectors of the studied area. In the lower part of the Sarah Fm, a claystone-dominated section from Well-5 to 7 is interpreted as lacustrine deposition based on the palynology assemblages. A core cut in Well-7 belongs to such section and mostly consists of micaceous claystones and a few examples of cm-thick, ripple-laminated sandstone layers. These lithologies do not contradict the lacustrine interpretation from palynology and suggest that deposition would have occurred in comparatively distal lacustrine settings relative to the sediment source area. A core cut in Well-6 is located in the upper part of the Sarah Fm and mainly includes m-thick bedsets of medium-grained sandstones with intense deformation (e.g., irregular folding and oversteepened bedding). Near the middle of the core, cm-thick black claystones with intense deformation by shearing are also present. The predominance of medium-grained lithologies points to a depositional site that often had relatively high energy conditions. Whether this was continental (fluvial) or subaqueous (shallow or deep marine) is difficult to constrain, since the core is intensely deformed and offers no clear criteria for sedimentological interpretations. Yet, our chronostratigraphy indicates that this interval belongs to the glaciogenic Sarah Fm and the observed intense deformation seems easy to reconcile with the common documentation of similar deformation in the Sarah Fm elsewhere in the Arabian Plate (Melvin, 2018; Craigie et al., 2016; Tofaif et al., 2018). An explanation would be that the intense deformation was related to braided fluvial or submarine fan strata that after deposition experienced intense loading by grounded glacial ice. An alternative is that glaciogenic submarine fans partly experienced intense deformation due to fluidization of the strata related to rapid deposition and loading by the overlying sediments. The available dataset, however, does not allow to discriminate between these two interpretations. Regardless, despite the intense deformation, compaction and cementation are not

extreme in these sediments and the implication is that some open pore spaces may still be present in the Sarah Fm across parts of the RAK Basin.

Lower to Middle Silurian

Based on our palynological data, the only Silurian strata present in the studied area correspond to a relatively thin section (50-300 ft thick) of the Qusaiba Member (i.e., the lower unit of the Qalibah Fm) in Well-5 to 7 (central and southern sectors). Similar to lower Silurian (Llandovery) strata elsewhere in the margins of Gondwana, deposition of the Qusaiba Member in the RAK Basin must have been related to sea-level rise after the Hirnantian glaciation, which promoted the formation of organic-rich sediments at least during the early stages of deposition. The high organic content often found in the lowermost interval of the Qusaiba Member is one of the proposed prerequisites for the fact that this interval is frequently considered a world-class source rock in the Arabian Plate (e.g., [Hayton et al., 2016](#)). Well-5 to 7 exhibit high GR-log values at the very base of the Qusaiba Member, just above the contact with the Sarah Fm, and these values may be related to an increased content in organic matter. The current burial depth of the Paleozoic strata in the studied sectors of the RAK Basin seems deep enough to have facilitated the maturation of any organic matter content and, together with the above situation, it highlights the relevance of the lowermost Qusaiba Member in this basin.

Above the interval with elevated GR-log values, Well-5 and 7 include cores within the Qusaiba Member that, following the regional trends, show lithologies with apparently no organic richness. These cores are dominated by micaceous claystones with finely-laminated fabrics that consist of abundant mm-thick siltstone to very fine-grained sandstone layers that show cross laminations and a few occurrences of horizontal trace fossils. These sediments are interpreted to belong to marine offshore settings that resulted from the rising sea level after the Hirnantian glaciation. Regardless, it is important to remark that our well correlations propose a limited presence of the Qusaiba Member in the studied area (only Well-5 to 7). Instead of non-deposition, such a limited presence may be a consequence of the foremost erosional event during the late Carboniferous and the subsequent formation of the Hercynian Unconformity, which many publications propose was due to a period of structural inversion after the onset of collision between Gondwana and Eurasia ([Ruban et al., 2007](#); [Faqira et al., 2009](#)). In addition to affecting Silurian strata, this erosional event is interpreted here as the cause for the absence of any Devonian sediments across the eastern part of the RAK Basin. In other words, the Hercynian Unconformity in the eastern part of the RAK Basin incorporates a gap in the sedimentary record of at least ~110 Ma that separates the lower Silurian Qalibah Fm from the upper Carboniferous to Lower Permian Juwayl Fm (i.e., the lower part of the Unayzah Group).

Upper Carboniferous to Lower Permian

Above the Hercynian Unconformity, our palynological examinations and well correlations show significant thickness variations (20-600 ft thick) in the upper Carboniferous to Lower Permian Juwayl Fm, with thicker sections in Well-3, 6 and 7 than in the other wells. Previous works on the Juwayl Fm have highlighted the glaciogenic origin of its stratigraphy (i.e., the Ghazal and Jawb members) across the Arabian Plate ([Melvin and Norton, 2013](#)). Based on such origin, the thickness variations observed in the Juwayl Fm is probably related to differences in accommodation space, subsequently filled by glaciogenic strata, that resulted from the shaping of the topography by glacial ice advances and retreats; that is, the thicker sections of the Juwayl Fm in our wells could correspond to thicker, stacked, glacial-valley, sedimentary fills, whereas the thinner sections would represent thinner valley fills or valley fills that were partially eroded due glacial dynamics.

The glaciogenic interpretation of the Juwayl Fm is confirmed in our study by the sedimentological inferences from a core cut near the top of the Jawb Member (upper part of the Juwayl Fm) in Well-6. This core mainly consists of (1) fining-upward (coarse-to fine-grained) bedsets of moderately-sorted, pebbly, massive to cross-bedded sandstones in the lower part, (2) finely-laminated, very fine-grained, silty sandstones in the middle part, and (3) up to coarse-grained, massive sandstone bedsets in the upper part that

contain abundant floating lithoclasts and possible dropstones. These sediments are, respectively, interpreted as glaciogenic braided fluvial, distal lacustrine, and subaqueous fan (possibly lacustrine) deposits and would correspond to the later stages of glaciation during the earliest Permian. Compaction and cementation in this core are not low and suggest the absence of open pore spaces in the Jawb Member at least in the location of Well-6.

In the case of the overlying Nuayyim Fm (Lower Permian), its lower unit, i.e., the Wudayhi Member, is present across the entire studied area except in Well-1 (northernmost sector); yet, its upper unit, i.e., the Tinat Member, is found in all the studied wells. Relative sea-level rise after the end of the late Carboniferous-earliest Permian glaciation is thought to have produced a major transgression in the Arabian Plate which, subsequently, promoted the landward advance of the gross depositional environments and facies belts by the time of formation of the Wudayhi Member. The fact that the Wudayhi Member is absent in Well-1, though, supports the notion that the northernmost sector of the studied area possibly was a paleo-high that experienced little or no deposition during most of the Paleozoic. Unlike the Wudayhi Member, the entire studied area, and possibly most of the eastern part of the RAK Basin, exhibits similar thickness for the overlying Tinat Member and such a situation suggests a time of relative structural quiescence at least until the end of deposition of this unit in this part of the basin. Two different cores cut in the lower and upper parts of the Tinat Member are respectively available from Well-2 (northern sector) and Well-7 (southern sector). The core in Well-2 has normally-graded, massive to cross-bedded, pebbly sandstones and intercalations of very fine-grained, dolomitic sandstones and claystones with trace fossils, microbial laminations, and desiccation cracks. This core is interpreted as deposition in a costal intertidal flat that was crosscut by tidal channels. The core in Well-7 shows a lower section with ripple-laminated or low angle cross-bedded sandstones that include trace fossils and bioclastic debris which are assigned deposition in a shallow-marine setting. The middle part of the same core exhibits dolomitic siltstones and claystones with common presence of horizontal and concretionary microbial fabrics, in addition to several sharp surfaces of karstification and desiccation cracks. The middle part of this core probably indicates intertidal sedimentation punctuated by periods of subaerial exposure. Finally, the upper part of this core presents massive, red claystones with mm-scaled cracks (i.e., possible rootlets) that, along with the underlying cored strata, suggest a final depositional stage in a coastal flood plain under subaerial conditions that facilitated an intense oxidation of the sediments and the partial colonization by small plants. The latter is probably the local expression of the commonly documented paleosols near the top of the Tinat Member elsewhere in the eastern Arabian Plate (see e.g., [Polo et al., 2019](#)), which resulted from an increased climatic aridity by the end of deposition of the Unayzah Group and whose top marks the so-called pre-Khuff Unconformity below the upper Permian Khuff Fm.

Regardless, only a moderate amount of open pore spaces is predicted to occur in the coarser-grained sandstones from the cores of the Tinat Member. A broader implication is that, if similar coastal settings with marine influences are present across the eastern part of the RAK Basin during the deposition of the Tinat Member, the relevance of this member is compromised by the possible presence of alternating sandstone and claystone lithologies following the inferences from these cores.

Conclusions

Our study shows one of the few published integrations of Paleozoic palynological and sedimentological data from wells of the Rub' al-Khali Basin and it can be summarized as follow. Above a Neoproterozoic basement, the middle Cambrian Siq and Burj formation are identified and partially correspond to fluvial and coastal plain environments. The upper Cambrian to Middle Ordovician Saq Formation is found in most of the wells except in the northern sector of the studied area, the latter probably forming a local paleo-high that experienced little or no deposition during large parts of the Paleozoic. The Saq Formation, partially assigned to a fluvial system, is separated by a ~15 Ma unconformity from the overlying Middle to Upper Ordovician Qasim Formation, with only its upper members present in the studied wells. Above, a comparatively thick

section of the glaciogenic Sarah Formation is found in the wells from the central and southern sectors. Its lower interval is partially attributed to lacustrine environments, while the upper parts may be related to deposition in fluvial or marine settings. The found stratigraphy of the overlying Qalibah Formation (lower to middle Silurian) is relatively thin and incomplete probably due to the major erosion that resulted in the formation of the Hercynian Unconformity during the late Carboniferous. Some organic rich intervals at the base of the Qalibah Formation may still occur in our wells. The glaciogenic Juwayl Formation (upper Carboniferous to Lower Permian) overlies the Hercynian Unconformity and reveals variable thickness probably due to glacial ice dynamics. Finally, the Lower Permian Nuayyim Formation exhibits relatively constant thickness across the studied wells, which is interpreted as resulting from a period of persistent subsidence and fill of the accommodation space. Importantly, parts of the strata of the Nuayyim Formation are interpreted to result from a dynamic interaction between coastal and marine systems, without any presence of the aeolian settings that otherwise have often been documented in areas of the eastern Arabian Plate.

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